

Forensics Case Report: Comprehensive Browser Artifact Analysis

CASE ID: 001 - Web Activity & Attribution Reconstruction

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TOOL UTILIZED: Browser History Examiner (BHE)

1. Executive Summary

A digital forensic examination was conducted on a suspect's local machine to investigate web browsing activity, reconstruct user behavior, and establish definitive attribution. Utilizing Browser History Examiner (BHE), a live capture of local browser artifacts was executed.

The extraction successfully recovered over 27,000 distinct artifacts across multiple browsers, despite potential anti-forensic measures like history deletion. By correlating cached media, stored credentials, session tabs, and search queries, the investigation successfully rebuilt the suspect's timeline, confirmed identity attribution, and revealed actionable insights regarding user intent and potential data leakage.

2. Case Background & Scope

The objective of this examination was to reconstruct the digital footprint of a suspect to determine their online activities, motivations, and potential unauthorized data access.

Investigative Goals:

- Capture volatile and non-volatile browser data directly from the host machine.
- Parse SQLite databases and local storage to extract history, credentials, and cached media.
- Assess the forensic value of the recovered data to establish *mens rea* (intent) and absolute attribution to the suspect.

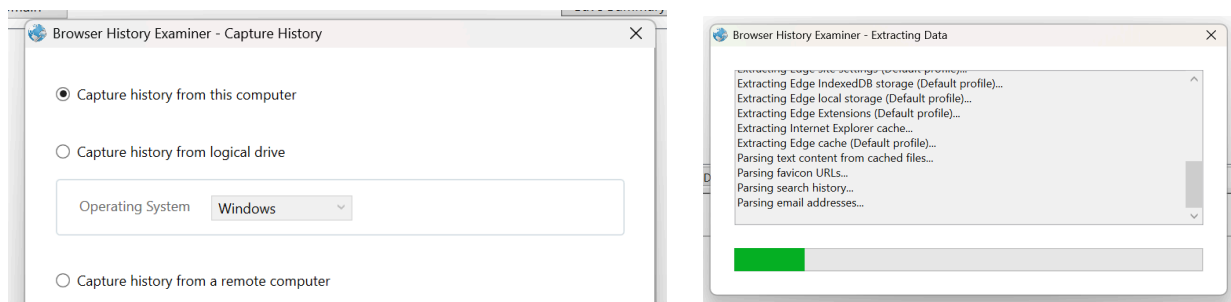
3. Evidence Acquisition

To ensure the integrity of the collected artifacts, data was acquired using a specialized forensic extraction protocol.

- **Extraction Tool:** Browser History Examiner (BHE)
- **Methodology:** Live System Capture

- **Procedure:**

1. **Tool Deployment:** BHE was launched directly on the target environment.
2. **Target Selection:** Initiated **Capture History -> Capture from this Computer**.
3. **Parameter Configuration:** Targeted specific user profiles across all installed web browsers (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge). A secure destination folder was designated for the captured image.
4. **Execution:** The capture process was executed, allowing the tool to parse live directories, SQLite databases (e.g., **History**, **Web Data**, **Cookies**), and hidden system cache folders.



4. Methodology & Artifact Analysis

Once the raw databases were acquired, the artifacts were loaded into the BHE analysis engine. Over 27,000 individual records were parsed, categorized, and examined. The findings were segmented into four key forensic domains: Navigation & Intent, Identity & Authentication, Local Storage & Cache, and System Configurations.

Browser History Examiner - Trial Mode						
File Options Filter Report Tools Help						
Artifact	Records	Bookmarks		Report Preview		
Bookmarks	75					
Browser Settings	14					
Cached Files	6351					
Cached Images	1282					
Cached Web Pages	66					
Cookies	118					
Downloads	633					
Email Addresses	151					
Extensions	6					
Favicons	7051					
Form History	1004					
Logins	162					
Searches	1560					
Session Tabs	211					
Site Settings	3716					
Site Storage	1749					
Thumbnails	20					
Website Visits	8713					
		Date Added	Last Modified	Title	URL	Web Browser (Profile)
		08/23/2025 16:09:12		AI Glasses, Sunglasses: Try Or	https://www.perfectcorp.com	Edge (Default)
		08/18/2025 12:02:34		How Pngme's digital credit sc	https://medium.com/pngme/	Edge (Default)
		08/18/2025 11:56:39		How to get Pngme e-float loc	https://infomediag.com/png	Edge (Default)
		08/18/2025 11:32:32		MTN AGENT LOANS - FAQs -	https://www.mtn.co.ug/help	Edge (Default)
		07/22/2025 10:29:33		Ghana Mobile Money Market	https://markwiderresearch.co	Edge (Default)
		07/22/2025 10:04:02		Value of MoMo transactions :	https://www.primenewsgh	Edge (Default)
		07/22/2025 09:54:37		17,000 MoMo businesses coll	https://www.myjoyonline.	https://markwiderresearch.com/ghana-mobile-money-market/
		07/22/2025 09:50:59		Liquidity for 'momo' agents -	https://www.graphic.com.gh/	Edge (Default)
		07/01/2025 09:57:01		Free Linux, DevOps & Cybers	https://labex.io/	Edge (Default)
		06/26/2025 10:35:21		INE - Penetration Testing Stuc	https://www.4funbox.com/sh	Edge (Default)
		06/11/2025 21:33:18		Free Events Leland	https://www.joinleland.com/e	Edge (Default)
		06/11/2025 11:18:57		Scholarships and funding St	https://study-uk.britishcounc	Edge (Default)
		06/01/2025 11:34:15		(44) Ethical Hacking in 15 Ho	https://www.youtube.com/wv	Edge (Default)
		06/01/2025 00:18:16		Tasks - Dashboard - Bugcrow	https://bugcrowd.com/dashb	Edge (Default)
		05/31/2025 15:11:26		University of Aberdeen - Ran	https://unisicholars.com/univ	Edge (Default)
		05/31/2025 11:31:19		*Rich Dad Poor Dad.pdf	file:///C:/Users/Max/Desktop	Edge (Default)
		05/31/2025 02:54:13		Course Introduction TCM Se	https://academy.tcm-sec.com	Edge (Default)
		05/25/2025 10:51:05		Module 2: Reconnaissance te	https://www.linkedin.com/lea	Edge (Default)
		04/24/2025 03:16:48		Read, write, or emulate NFC t	https://www.luisllamas.es/en/	Edge (Default)
		04/20/2025 21:52:56		DeepSeek Platform	https://platform.deepseek.co	Edge (Default)
		04/14/2025 18:22:52		IT Masters Short Courses	https://learn.itmasters.edu.au	Edge (Default)
		01/29/2025 13:32:33		Northern Arizona University	https://www.applyweb.com/c	Edge (Default)
		01/28/2025 23:49:37		GradCAS My Application Da	https://gradcas2025.liaisonca	Edge (Default)
		01/14/2025 18:18:05		Display Fundamentals - TFT	https://doc-tft-espi.readthed	Edge (Default)
		01/08/2025 20:11:25		Login Networking Academy	https://auth.netacad.com/aut	Edge (Default)

4.1. Navigation & Intent (Establishing Motive)

These artifacts formed the backbone of the chronological timeline, revealing exactly what the user was looking for and when.

- **Website Visits (8,713 records):** Core chronological browsing history containing URL structures and timestamps.
- **Searches (1,560 records):** Raw search engine queries providing direct insight into the suspect's mindset, planning, and immediate interests.
- **Form History (1,004 records):** Captured typed inputs (names, addresses, specific keywords) revealing deliberate thought processes.
- **Session Tabs (211 records):** Captured the exact state of the browser at the time of the last session, freezing real-time activity.
- **Bookmarks (75 records):** Indicated long-term interests and intentionally saved resources.

4.2. Identity & Attribution (Tying the User to the Keyboard)

To prove *who* was sitting at the computer, authentication and identity markers were extracted.

- **Logins (162 records):** Stored usernames and passwords directly tying the local user profile to remote online accounts.
- **Email Addresses (151 records):** Autofill and registration data linking the suspect's known aliases to the browsing activity.
- **Cookies (118 records):** Session tokens that proved active, authenticated access to specific web services.

4.3. Local Storage, Media & Cache (Recovering Hidden Evidence)

Crucially, even if a user deletes their primary history, the browser cache retains massive amounts of residual data.

- **Favicons (7,051 records):** Website icons that act as "residual traces." Even when the main history database is wiped, favicons often remain, proving a domain was visited.
- **Cached Files (6,351 records):** Locally stored temporary web elements. Highly useful for recovering traces of deleted browsing evidence.
- **Site Storage (1,749 records):** Persistent web app data, including IndexedDB entries and potential chat logs from web-based messengers.
- **Cached Images (1,282 records) & Thumbnails (20 records):** Visual artifacts confirming the suspect's exposure to specific media, images, or documents.
- **Cached Web Pages (66 records):** Allowed for the exact visual reconstruction of what the suspect saw on their screen, regardless of whether the live website was later altered or taken down.

4.4. System Behavior & Data Leakage

- **Site Settings (3,716 records):** Indicated site-specific permissions (e.g., camera, microphone, notification access), identifying trusted domains.
- **Downloads (633 records):** Documented files retrieved from the web, allowing investigators to tie web activity directly to files found on the local hard drive (vital for proving data exfiltration/leakage).
- **Browser Settings (14 records) & Extensions (6 records):** Revealed search engine preferences, privacy configurations, and installed add-ons (such as VPNs, password managers, or crypto wallets) indicating attempts at operational security or anonymity.

5. Conclusion & Forensic Implications

This examination demonstrated the sheer volume of actionable intelligence retained by modern web browsers. The forensic value of this BHE capture was absolute across several domains:

1. **Attribution:** By cross-referencing the 162 stored logins and 151 email addresses with the active session cookies, the browsing activity was definitively attributed to the suspect, eliminating the "malware defense" or claims of a shared workstation.
2. **Reconstruction of Intent:** The combination of 1,560 search queries and typed form history painted a clear picture of the suspect's motivations and planning prior to the incident.
3. **Evidence Resiliency:** Despite common operational security measures employed by suspects (e.g., VPN usage or clearing the primary history tab), the extraction of nearly 15,000 cached files, favicons, and site storage records provided a durable, secondary timeline of events.

Ultimately, this analysis reinforces that local browser artifacts, when properly acquired and parsed, remain one of the most reliable and comprehensive sources of digital evidence for reconstructing user behavior.